

Scoring Students' Answers Using OCR and GPT

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Abstract—Artificial intelligence (AI) has immense potential to revolutionize grading and the delivery of efficient feedback to students, all while reducing the workload for instructors. Existing automated grading systems are typically limited to specific question types, such as true-false and multiple-choice questions, and often do not address the need to assess more diverse forms of student responses. This paper proposes ExaGrade, an open-source platform that integrates optical character recognition (OCR) with the Generative Pre-trained Transformer (GPT). It enables the use of AI models to automate the grading of student answers. The platform supports both paper-based and online exams. The early results presented in this paper show promising capabilities in handling flexible formats for true/false and multiple-choice questions. However, further investigation and refinement are necessary to effectively address open-ended responses. The study observed that GPT's scoring of student answers faces notable limitations compared to instructor evaluations. The experiment revealed a low average Euclidean distance, indicating significant differences between the two grading methods. While the average cosine similarity of 0.628 suggests moderate alignment in grading patterns, the average exact match ratio of 0.648 reflects a moderate overall agreement between ExaGrade and the instructor's evaluations.

Index Terms—Artificial Intelligence (AI), Automated Scoring, Optical Character Recognition(OCR), Generative Pre-trained Transformer (GPT).

I. INTRODUCTION

Automated essay scoring (AES) systems aim to evaluate the quality of essays using computer-based technology. They are gaining growing attention in the field of education and information technologies due to their ability to assess student responses more quickly and efficiently. Manual evaluation of writing, particularly for open questions such as essays, problem solving tasks, and handwritten responses, is time-consuming and labor intensive. Recent advances in artificial intelligence (AI), natural language processing (NLP), and machine learning (ML) provide the potential to automate the assessment of a wider variety of question types with improved accuracy and consistency; however, offered technologies currently fall short of meeting various needs of teachers and students in real-world teaching scenarios[1].

Large language models (LLMs) such as Generative Pre-trained Transformer (GPT) [2] are advanced AI systems characterized by their vast number of parameters and exceptional learning abilities[3]. At the heart of many LLMs is the self-attention mechanism found in the Transformer [4] architecture, which transformed the field of NLP by efficiently handling

sequential data, enabling parallelization, and effectively capturing long-range dependencies in text. ChatGPT[5], which is a chatbot application that uses the GPT model to engage in conversations, has opened new opportunities for millions daily active users in various tasks, including information retrieval, writing assistance, studying, planning, and casual conversation in personal, academic, and professional contexts.

II. RELATED WORK

This section discusses AES, specifically its evolution within the context of LLMs. Then, it investigates the application of OCR technology in the processing of student response documents, assesses various contemporary online tools for test scoring, and concludes with an analysis of the leading approaches in automatic evaluation.

A. AES in the Era of LLMs

AES is no longer limited to surface-level features such as word count and grammar mistakes; modern AES systems now harness transformer-based models to provide nuanced evaluations of textual coherence, argumentation, and stylistic quality [6]. State-of-the-art AES systems are predominantly built on supervised learning models and commonly address three core tasks: (1) regression, aimed at predicting a score for an essay; (2) classification, which categorizes essays into predefined levels such as low, medium, or high quality; and (3) ranking, which compares and orders essays based on their relative quality [7].

Research on GPT models has been conducted to evaluate exam scores. A fine-tuned GPT-2 that is enhanced with domain-specific data rivals traditional AES systems in both validity and reliability, indicating its potential to be a scalable scoring tool in various educational settings [8]. GPT-3 has effectively scored 12,100 essays from a TOEFL dataset, authored by non-native English speakers, and showed a strong correlation with human scores [6].

However, there are studies that show limitations of ChatGPT as an AES tool [9], [10], [11]. ChatGPT has been explored as a tool for grading writing quality in writing classrooms, compared to an experienced human rater[9]. The results indicate that ChatGPT often scores lower and its evaluation is inconsistent due to variations in ChatGPT's scoring mechanism, training data, model updates, and ingrained randomness. Additionally, the consistency of ChatGPT's scores over multiple

evaluations was low, indicating variability and raising concerns about its reliability.

B. OCR for Handwritten Answers

OCR has been employed to digitize both handwritten and printed texts. In educational settings, OCR provides an effective means of converting scanned or photographed student work such as assignments, test papers, and handwritten notes into machine-readable formats. The digitization of these documents streamlines academic workflows and also facilitates the automation of student answers grading. Multiple-choice questions (MCQs) and true/false (TF) answers can be graded using MCQ evaluation as in [12], or through optical mark recognition (OMR) scanners that follow a predefined template. Automatic grading of MCQs and TFs requires using standardized symbols, pre-formatted answer sheets, or strictly aligned answer boxes [13].

There is a wide array of both open-source and commercial OCR engines available, each offering unique advantages and capabilities. Tesseract is a prominent open-source OCR engine, which is currently maintained by Google [14]. Tesseract incorporates deep learning-based models (LSTM), supports over 100 languages, and is both extensible and cross-platform, making it widely adopted in academic and industrial projects. There are other open-source alternatives include OCRopus[15], Calamari[16], and EasyOCR[17]. Commercial OCR engines, which are often cloud-based, are designed to meet enterprise-level demands by providing robust and scalable solutions. Services such as Google Cloud Vision OCR [18], Amazon Textract [19], and HandwritingOCR[20] offer recognition of printed and handwritten text, as well as structured content like form fields and tabular data. These cloud-based OCR systems are typically accessed through RESTful APIs, enabling seamless integration with web and mobile applications. In contrast, open-source OCR engines are cost-effective libraries that are integrated directly into the hosting web server.

C. Online Test Scoring Tools

Online test grading applications streamline the assessment process by transforming smartphones or tablets into optical grading scanners. These tools allow educators to efficiently grade multiple-choice answer sheets using the device's camera, providing immediate feedback and detailed performance analytics. Several affordable solutions are available, including ZipGrade[21], GradeCam[22], and Akindi[23]. These platforms offer similar capabilities, such as support for customizable answer sheets, multiple test versions, and integration with student rosters. However, their grading functionality is limited to score MCQ and TF question types.

More advanced test scoring tools leverage AI to improve grading efficiency and consistency. In Gradescope[24], the instructor identifies the question and answer regions on a few sample sheets, after which the AI generalizes the grades provided and feedback across similar student responses to

streamline the grading process. Cognii[25] offers an assessment engine that employs data mining and machine learning techniques to improve the accuracy of scoring in essay comprehension tasks. A controlled research study reported that Cognii's evaluations achieved a 96% agreement rate with human grading of short essays[25]. However, it remains unclear whether Cognii is suitable for assessing broader types of questions.

D. Automatic Scoring Approaches

Several approaches and systems have been proposed in the field of automatic scoring and grading. Many of these methods focused primarily on single techniques, such as essays, short answers, or multiple-choice questions. In addition, most papers assume digital-format answers.

There are approaches that focus exclusively on MCQs without the need for pre-formatted answer sheets. The paper [13] uses Tesseract as an OCR engine and YOLOv8[26] as an object detection engine. The goal of their approach is to simplify test processing for both educators and students. The authors identified some false positives as a limitation in their MCQ evaluation system. Their results confirm our findings regarding Tesseract's limited performance as a cloud OCR engine. The paper [27] employees a deep learning approach based on SAST[28] model, which balances accuracy and speed, able to detect text of different shapes, and show multilingual generalization. D. Tardioli, F. Aznar, and A. R. Mosteo introduced an innovative method for students to select answers using 2D barcodes (QR or Aztec codes), notably requiring minimal technology[29].

H. Balaha and M. Saafan proposed an automatic exam correction framework called HMB-AECF to correct the MCQs, essays, and equations[30]. HMB-AECF is structured into five distinct layers: question and student answer combination layer, student and reference answers combination layer, auto correction layer, manual correction enhancement layer, and results output layer. To give a mark for an essay questions, the framework measures the similarity between texts using methods such as Cosine similarity, Jaccard similarity, Manhattan distance, and Euclidean distance. The proposed framework places a high emphasis on grading of mathematical equations, and it employs a specific algorithm to assess the similarity between student-answer equations. While ChatGPT and other AI models are capable of functioning as assessment engines within HMB-AECF, the framework does not support paper-based submissions.

The method developed by V. Rowtula, S. R. Oota, and C. Jawahar focuses on word spotting in handwritten answers, distinguishing it from full-text transcription without character-level segmentation [31]. Their approach integrates convolutional neural networks (CNN) for feature extraction and self-supervised learning to adapt to diverse questions. However, their methodology is centered on narrative-type questions, which limits its generalizability across different question types.

The field of automatic grade and feedback tools (AGFTM) is experiencing rapid development, driven by the increased

demand for efficient evaluation of open-ended responses and essays that require contextual understanding. Experiments with machine learning algorithms as evaluation tools demonstrate precise and contextually aware grading [32]. Artificial neural networks (ANNs), particularly LSTM, is mostly applied in the research, while similarity-based methods, such as cosine similarity and graph similarity, have been used in various papers[33].

III. PROPOSED WORK

This paper presents ExaGrade, an automatic grading and feedback tool built to integrate and leverage AI models. Its core purpose is to investigate how AI can reduce the manual burden on educators while simultaneously boosting the accuracy, consistency, and speed of grading across both objective and subjective question types. ExaGrade automates the exam grading process both digital and paper-based tests using two key technologies: OCR for recognizing handwritten student responses, powered by HandwritingOCR, and GPT-4 for automated grading. Based on experiments, HandwritingOCR demonstrated superior performance in converting paper-based student exams compared to Tesseract and other open-source Python libraries, making it the preferred choice. The proposed system supports multiple question types, including MCQs, TFs, short- and long-responses questions.

The following two subsections will discuss the activities involve in ExaGrade and the tool implementation.

A. ExaGrade Workflow

The workflow of ExaGrade for both online and paper-based exams is illustrated in Fig. 1. The grading process in ExaGrade involves the following key steps:

- **Exam Creation:** The instructor of the course begins by creating an exam for his/course that may incorporate various types of questions. During this stage, the instructor defines the exam model, which outlines the structure, scoring, and parameters for each type of question. Once the exam is created, the instructor proceeds to solve the exam himself. This crucial step generates the answer model, which serves as the ground truth for grading. For MCQs and TFs, the instructor provides the correct responses; for short- and long-responses questions, the instructor outlines the expected content.
- **Student Exam Completion:** Students complete the exam using traditional paper-based formats, writing their answers by hand as usual. Alternatively, the exam may be administered online, in which case it will not undergo scanning and OCR text extraction.
- **Scanning Exam Papers:** After students complete the exams, the instructor collects all the papers and scans them. These scanned images are then fed into the ExaGrade system.
- **OCR Text Extraction:** ExaGrade takes each scanned exam paper and sends it to HandwritingOCR through its API. HandwritingOCR analyzes the image of the paper, detects the handwritten, and converts it into a digital format. This

extracted text represents the student's answers to each question. The RESTful API call for HandwritingOCR can be found in `extract_text_from_pdf()` method inside the file 'ExaGrade/utlis/utlis.py'.

- **AI Answer Scoring:** Finally, the extracted text from each exam paper is sent to GPT-4. Using the answer model provided by the instructor, evaluates each question. For objective questions (MCQs, TFs), it matches the extracted answer against the correct option. For subjective questions (short and long answers), ChatGPT assesses the content, coherence, and accuracy of the student's response against the expected answers and grading criteria, assigning a mark or score accordingly.

ExaGrade provides flexible evaluation criteria for long and short question types. Therefore, instructors can establish clear guidelines for evaluating student responses, balancing strictness with flexibility depending on the nature of the task. The system offers five evaluation criteria:

- **Strict:** Grading is based on an exact match with the solution, such as for numerical problems or specific coding syntax. No credit is given for any deviation from the correct answer.
- **Flexible:** Accepts answers that convey the same meaning but may be differently phrased. In this case, a student may receive a partial credit if the answer is close but misses important information.
- **Keywords:** Checks for the presence of key concepts from the solution in the student's answer. Partial credit can be awarded if relevant keywords are present, even without explicit naming. Credit can vary from full for perfect solutions to partial for minor errors.
- **Coding:** Evaluates logic, syntax, and correctness of code output.
- **Custom:** Allows judgment based on instructor criteria. It provides instructors with the flexibility to set unique and question-specific criteria.

Listing 1 presents the GPT prompt utilized in ExaGrade, sourced from "utlis/AI_utlis.py". This code has been slightly modified for improved readability. The prompt will be complete after including the variables: question, eval_type, solution_text and the student answer in student_text.

```
...
eval_instructions = {
    "strict": "Grade strictly based on exact match with the solution.",
    "flexible": "Accept answers that have the same meaning, even if phrased differently.",
    "keywords": "Check whether key concepts from the solution are present in the student's answer.",
    "coding": "Grade this as a code answer. Check logic, syntax, and correctness of the output.",
    "custom": custom_note.strip() or "Use your judgment to grade this based on the instructor's custom criteria.",
}
prompt = f"""
```

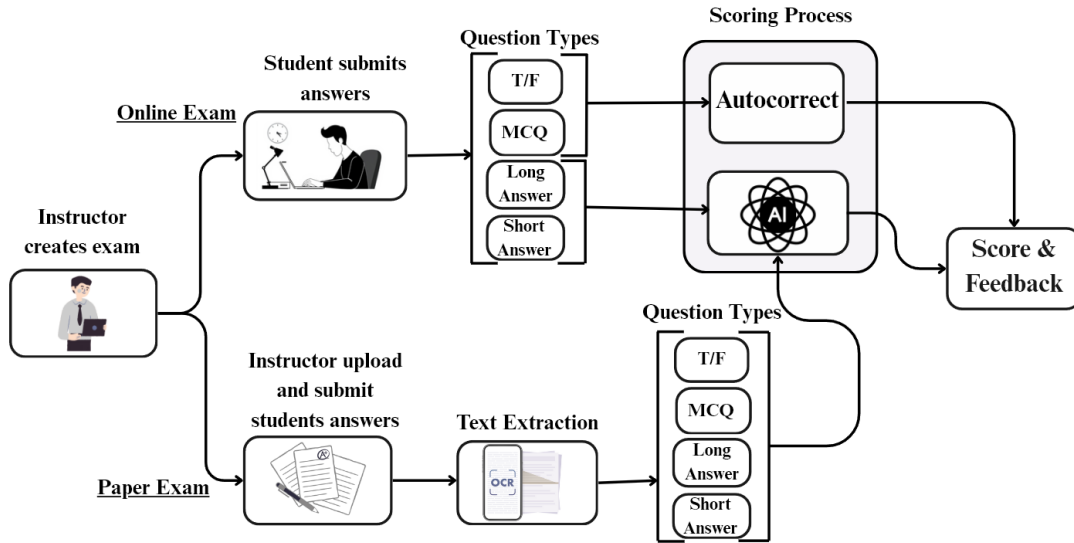


Fig. 1. The workflow for grading online and paper-based exams in ExaGrade.

```

You are an AI grading assistant for {question_type.
upper()} questions.
IMPORTANT:
- Only consider the first clear answer given by
  the student.
- Ignore any explanations, comments, or sentences
  after the first answer.
- Match as follows:
  - For True/False:
    - "T", "True", "✓", "√" → True
    - "F", "False", "×", "X" → False
  - For MCQ:
    - "A", "B", "C", or "D" (case insensitive)
- If multiple answers (e.g., "A and B") appear in
  the first few words, mark it ⚠️ REQUIRES
  ATTENTION.
- If answer is unclear or missing, also mark it ⚠️
  REQUIRES ATTENTION.
- Do not overthink explanations. Just match the
  answer against the solution.
Return your output ONLY in this strict format:
Score: x/{marks}
Feedback: Correct ✓, Incorrect ✗, or ⚠️ REQUIRES
ATTENTION.

Question: {question}
Student Answer: {student_text}
Solution Text: {solution_text}
"".strip()
...
  
```

Listing 1. Excerpt from grading.py displays the GPT prompt.

B. Platform Implementation

To enhance accessibility and usability, the ExaGrade platform is released as an open-source project, available on GitHub [34]. It enables educators to create exams, upload exam papers, receive grades, and review feedback through a user-friendly online interface. The system is developed using Django framework. Detailed instructions for installing local version, along with the required technologies including the API keys (for GPT and HandwritingOCR) are provided in the ExaGrade repository.

IV. EXPERIMENTS AND RESULTS

The capabilities of GPT-4 and HandwritingOCR within ExaGrade is used to validate the scoring of a typical exam in an intermediate Java programming course within an academic environment. The dataset consists of handwritten exam responses from 13 students. The exam featured seven questions: five TF questions and two long-answer. Fig. 2 shows a sample of the exam. All submissions were scanned and uploaded to the ExaGrade as well as the answer model and the max scores.

The performance of ExaGrade was compared with that of an experienced instructor using the evaluation metrics defined in Table I. These metrics quantify the similarity and agreement between the system-generated and human-assigned scores. Specifically, the Average Euclidean distance measures the overall magnitude of deviation between corresponding scores, with smaller values indicating closer agreement. The Average Cosine similarity captures the directional consistency between score distributions, assessing whether ExaGrade's grading pattern aligns with the instructor's. Finally, the Average (Mean) metric computes the mean score across all responses, providing a general measure of grading tendency or bias.

The following two subsections present the results of the OCR evaluation and ExaGrade's scoring.

A. OCR Evaluation

The first stage of the exam answer processing involved extracting student responses from scanned exam papers using the Handwriting OCR API. Tesseract OCR did not achieve optimal results for this purpose. Our experiments with Tesseract can be found in the "dataset/OCR" directory. Using Handwriting OCR, the system successfully recognized all student responses on the seven questions, achieving excellent character recognition accuracy. Notably, even answers written outside

■ True/False Questions

* Write 'True' or 'False' clearly in the box.

1. Instantiate an object of an abstract class causes an error. [1 marks]
2. An abstract must have at least one abstract method. [1 marks]
3. Any child class must implement all abstract methods of the abstract parent class. [1 marks]
4. Downcasting of an object is performed implicitly. [1 marks]
5. One interface in Java can have more than one father. [1 marks]

True

True

False

True

False

■ Long Answer Questions

* Write your detailed response below.

6. Use code example to explain the concept of polymorphism in Java. [2 marks]

```
class Animal
class Animal extends Horse
class Animal = new Horse()
```

7. Implement an interface "QuizInterface" that has an int method getGrade(). Then, implement a class Quiz3 that uses the interface "QuizInterface". [3 marks]

```
class QuizInterface = getGrade()
class Quiz3 extends QuizInterface
class QuizInterface = getGrade()
class Quiz3 = QuizInterface
```

Fig. 2. A sample of the programming exam (quiz), which was answered by a student.

TABLE I
DEFINITIONS OF METRICS

Metric	Definition
Average Euclidean	$\text{Avg. Euclidean} = \frac{1}{N} \sum_{i=1}^N \sqrt{\sum_{j=1}^d (x_{ij} - y_{ij})^2}$
Average Cosine	$\text{Avg. Cosine} = \frac{1}{N} \sum_{i=1}^N \frac{x_i \cdot y_i}{\ x_i\ \ y_i\ }$
Average (Mean)	$\text{Avg} = \frac{1}{N} \sum_{i=1}^N v_i$

predefined bounding boxes were accurately detected, highlighting the robustness of the recognition against handwriting variability and layout inconsistency.

B. Answer Scoring

Table II shows the instructor score and the ExaGrade score for the student sample "001". The dataset containing scores for all students in addition to the original submission files can be found in "ExaGrade/dataset". Table III shows the scoring for all samples; "Ins. Obj." denotes the instructor's scores for objective questions (Q1–Q5), while "Ins. Subj." denotes scores for subjective questions (Q6–Q7). The ExaGrade columns follow the same structure. The difference columns report the deviation between instructor and ExaGrade scores for objective, subjective, and total marks.

The results of the experiment show moderate agreement between the instructor and ExaGrade in grading individual

TABLE II
SCORE SUMMARY FOR A STUDENT SAMPLE ANSWER

Question	Max Points	Instructor Score	ExaGrade Score
Q1	1	1	1
Q2	1	0	0
Q3	1	1	1
Q4	1	0	0
Q4	1	0	0
Q6	2	1	0
Q7	3	1	0

responses. As shown in Table IV, there is a notable difference in the average Euclidean similarity score, indicating that one corrector consistently graded higher or lower than the other. Additionally, the low average cosine similarity suggests frequent disagreement on which questions deserved more points. The Average Exact Match Ratio reveals that 65% of the question scores matched exactly.

After a careful review of the scoring of ExaGrade's, here are our remarks:

- Scoring TF Questions: ExaGrade demonstrated strong consistency in grading objective answers, aligning with the instructor's evaluation in 11 out of 13 cases. This indicates its effectiveness in scoring structured, clear-cut questions. A key strength of ExaGrade is the ability to detect variations in True/False answers, such as "True", "T", a check mark (✓), "False", "F", or a cross mark (X).
- Scoring Long Answer Questions: We observed that the instructor tends to award partial credit generously, even for incomplete responses. In contrast, ExaGrade applies stricter grading even when the evaluation criteria is relaxed to "flexible". For instance, ExaGrade struggled significantly with Question 6: "Use a code example to explain the concept of polymorphism in Java". The answer involves programming syntax, code structure, and logical reasoning. ExaGrade assigned a score of zero to all students, including those who provided reasonably acceptable answers.

V. DISCUSSION

The experimental results validate the utility of the ExaGrade open-source platform in automating the grading of objective and highly structured question formats. The observed accuracy in these cases is attributable to the constrained answer space and explicit matching criteria that characterize well-defined questions. This success suggests that AI grading platforms can significantly reduce the administrative load associated with standardized assessments.

The study revealed a significant divergence in scoring accuracy when the GPT model was applied to open-ended, domain-specific questions. These discrepancies underscore the current limitations of black-box AI models in evaluating nuance and context. Specifically, the model struggled to provide equitable partial credit, a pedagogical necessity that often requires subtle contextual understanding and inference of student intent—qualities that go beyond simple semantic matching. This

TABLE III
COMPARISON OF INSTRUCTOR AND EXAGRADE SCORES

Sample	1-Ins. Obj.	2-Ins. Subj.	3-Ins. Total	4-Exa. Obj.	5-Exa. Subj.	6-Exa. Total	Diff. 1&4	Diff. 2& 5	Diff. 3&6
1	2	2	4	2	0	2	0	2	2
2	4	4.5	8.5	1	0	1	3	4.5	7.5
3	2	3.5	5.5	2	0	2	0	3.5	3.5
4	3	3.5	6.5	4	0	4	-1	3.5	2.5
5	2	3.5	5.5	1	1	2	1	2.5	3.5
6	4	5	9	4	1	5	0	4	4
7	2	3.5	5.5	2	0	2	0	3.5	3.5
8	2	3.5	5.5	2	1	3	0	2.5	2.5
9	2	4.5	6.5	1	1	2	1	3.5	4.5
10	3	4	7	3	0	3	0	4	4
11	4	5	9	4	2	6	0	3	3
12	4	4.5	8.5	4	0	4	0	4.5	4.5
13	2	2	4	2	0	2	0	2	2

TABLE IV
SIMILARITY METRICS BETWEEN GRADING OF EXAGRADE AND INSTRUCTOR

Metric	Value	Interpretation
Avg. Euclidean	0.293	Significant differences.
Avg. Cosine	0.628	Moderate-to-low agreement on grading patterns.
Avg. Exact Match Ratio	0.648	Indicates moderate agreement.

challenge is compounded in domain-specific fields where specialized terminology and conceptual relationships are critical to accurate evaluation.

The observed differences between AI and human grading highlight two critical areas for improvement. First, there is an immediate need to refine grading rubrics for greater clarity and consistency to better operationalize the criteria for partial credit. A more granular rubric serves as a clearer instructional signal for the AI model. Second, the system’s performance suggests that calibration of GPT’s grading behavior under direct instructor guidance is essential. This calibration should move beyond mere fine-tuning and incorporate a human-in-the-loop validation process to align the system with individualized, disciplinary grading expectations and ensure the ethical assignment of credit.

VI. CONCLUSION

This paper presented ExaGrade, a platform that integrates HandwritingOCR to extract text from handwritten exam papers and employs GPT-4 to evaluate and score student responses.

ExaGrade was tested on handwritten submissions from a Java programming exam. The system successfully converted paper-based answers into digital text while maintaining flexibility in student expression. For example, a “True” response may appear as “True,” “T,” or a check mark (✓), with similar variations for “False.” This design removes the rigid constraints of traditional bubble sheet formats.

In scoring, ExaGrade achieved strong performance on TF and MCQ format but showed limitations in grading long-answer questions requiring code comprehension and partial credit evaluation.

VII. FUTURE DIRECTIONS

This study provides an initial assessment of ExaGrade’s capabilities. Further work is needed to expand and validate its performance across broader datasets and grading contexts.

Future efforts will focus on larger, more diverse datasets covering multiple subjects and instructor evaluation styles. We also plan to test alternative LLMs, such as GPT-5 and omni models like GPT-o3, to compare performance across different question types, including coding tasks.

Additionally, the study will involve multiple instructors in grading the same exam papers, as human evaluation—especially for subjective responses—can vary considerably. This variability should be analyzed and controlled to better benchmark ExaGrade’s performance against human graders.

Finally, we aim to explore the integration of instructor-provided sample corrections to calibrate ExaGrade’s grading process and improve alignment with individual grading styles.

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